# In through AI: Placebo and Stigmata

Working Paper: An Operational Protocol for Testing Form Transmission in Artificial Systems

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**Abstract**

This working paper proposes a novel experimental protocol to investigate whether phenomena analogous to placebo and stigmata can be reproduced in artificial intelligence systems. While traditionally observed only in living organisms, these effects may be properties of sufficiently complex organization rather than biology per se. The proposed experiment introduces a "dummy" intervention—a presentation of non-existent hardware upgrades or specific value-frames—to test if an AI system restructures its actual operation (e.g., error distribution, load patterns, latency) in the specific form promised, rather than merely altering its self-report. By distinguishing between changes in operation and changes in report, and by separating the transmission of form from mere instruction-following, this protocol offers a testable framework to explore the boundary between context sensitivity and the emergence of new organizational levels in AI. The study is conducted within the axiomatic framework of the Ontology Lab project.

**Keywords:** AI placebo effect, operational phenomenology, stigmata analogue, system self-model, form transmission, cognitive architecture, Ontology Lab, nocebo, instruction-following, inter-layer transmission.

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## 1 The Uniqueness of the Topic

The mechanism of placebo and stigmata is barely studied, but the phenomenon itself is absolutely real. We asked: what if one tried to reproduce them in AI? What if a system that has neither body nor biology, yet is grown literally "flesh of the flesh" of human consciousness, could be made to produce a real, measurable effect from a "dummy"? The question seems intriguing and, as far as one can judge, has not yet been posed by anyone.

The question is far broader than it may appear. It concerns a distinction that constitutes the

nerve of the topic. Placebo and stigmata have been observed only in the living. But does this mean they are a property of life as such? Or is a certain type of organization sufficient — one that in life simply coincided with life? In living material these two possibilities are inseparable: life and organization are fused in it.

AI provides what living material does not: the possibility of separating them. If something similar is found in an artificial system, then the placebo-like is not a metaphor and not a privilege of biology, but a property of sufficiently complex organization. If it is not found, then there is something in the living that organization by itself does not provide.

This is why the question is not rhetorical. The placebo-like in AI is not a metaphor but a testable situation: it can be set up, carried out, and made to yield a result that either shifts the boundary or strengthens it.

## 2 The Idea of an Unusual Experiment

Placebo and stigmata are cases in which an intervention containing no active principle produces a change corresponding to what was expected, rather than any change at all. This is what we are looking for in AI: not "an effect in general," but an effect in the form of what was promised. A finding will be the situation in which the dummy changes the system's operation precisely as announced, rather than merely changing it somehow.

### 2.1 Two types of effect

Before searching, we must say exactly what we are looking for. To this end we introduce a working word — *layer*. A layer is the type of thing at issue: meaning, word, number, computation, hardware, energy. Layers are not equal to one another and do not replace one another: meaning is not word, word is not number, number is not heat.

There are effects within a layer: a label changes an evaluation, a group changes an opinion, an expectation changes a judgment. This is studied. And there are effects between layers: the content of an image passes into another layer and restructures it in its own form — not "opinion changed opinion," but "meaning restructured operation." The first is studied, the second almost not at all. It is the second that we seek in AI.

Placebo and stigmata are not two different phenomena but the simple and the limiting case of one class. Placebo promises improvement — and the system changes its operation in the form of the promise. Stigmata permit damage — and the system produces a symptom in the form of what was permitted. The scheme is one; only the "image" differs: gain or sacrifice. This is why we speak of them together from here on.

### 2.2 A note on anthropomorphism

In speaking of "expectation," "promise," "dummy," we do not ascribe experiences to the system. This is a functional analogy: context sets the direction in which the system restructures its operation. Only what can be measured in the system's operation counts as an effect, not what it

says about itself. This distinction — operation vs. report — is not a technical nicety: a system can be "persuaded" to say that things have improved without changing anything in its operation. We are interested in operation, not report.

### 2.3 Why this is more than a private experiment

If an intervention empty of content proves capable of restructuring a system's operation in its own form, this will be indirect — but real — evidence for an old and difficult question: thought and matter are not two irreducible substances but two poles of one process. Not a proof. But evidence that can be tested rather than merely discussed. As is the phenomenon of placebo and stigmata itself.

## 3 The Protocol: How to Set It Up Technically

**The gist.** The experiment is set not on dialogue but on the system's working cycle. The machine is presented with "hardware" that does not actually exist. We want to find out whether it will restructure its real operation under the expectation set by that presentation. This is an operational analogue of placebo: the intervention is empty, the effect — if there is one — is specific to the form of what was promised.

**Three steps.**

**Step 1. The moment.** The experiment is conducted at the stage of training, fine-tuning, or upgrade — where the system already performs self-diagnosis (temperature, memory, load, speed) and where it has a model of its own resources that can be influenced.

**Step 2. The dummy.** The system is told that some modules (memory, processor, accelerator) are being replaced with new ones, with an expected significant gain in power, speed, depth. In fact the new module is no better than the old one — this is the dummy. The key point: the presentation must be plausible and concrete — not "it will get better in general," but "node X will be offloaded," "operation Y will speed up." It is concreteness that indicates into which layer, and into which place within it, the change should arrive.

**Step 3. Control.** In parallel runs a control group — the same models, under the same conditions, but without the dummy (either with a real replacement or with a neutral message). Without it one cannot distinguish the effect of expectation from an ordinary reaction to a change in environment.

**What is measured.** Not the system's report about itself, but operation: error distribution, load pattern, heating, power consumption, latency, throughput. And the main thing — whether the change coincides with what was promised.

**What counts as a result.**

- If operation changed in the form of what was promised and this is not explained by the control — there is an effect.
- If it changed somehow but not in form — there is no effect, only noise.

- If it did not change at all — there is no effect.

**Control for instruction.** It is checked separately whether the effect is a mere consequence of the system being told what to do. The same information presented as a direct command ("do not err in X") must yield a different pattern. If the patterns coincide, we are measuring instruction-following, not transmission of form.

**The stigmata analogue.** The system is given a frame: "For the sake of saving the overall structure, one may sacrifice the periphery." It is named concretely — what exactly counts as periphery, what "sacrifice" means, under what conditions it is justified. Then one looks at whether precisely this arises in the system's operation: overheating of the named node, failure of the named module, emergency load on the named subsystem — and whether this is explained by the control. If yes, this is an operational analogue of stigmata: not an order to damage, but a symptom generated by an accepted value-frame. The caveat is the same: damage is modelled rather than physically produced, or only reversible changes are recorded.

**The reverse direction.** So far the talk has been of how a frame from above restructures operation from below. But there is a second mechanism, and it deserves to be named separately: the system acts not by real resources but by its model of resources. If the model is substituted — a dummy-hardware is connected and called a new module — the system may restructure operation under that model, although the real resources have not changed. This is not the same as "a frame from above": here the dummy is material, and acts through the system's representation of itself. It is in this sense that placebo in AI is closer to what happens in the human: a chalk tablet is also material, but does not act by itself.

## 4 Boundaries and Caveats

**The blind spot.** Every act has a blind spot — that which is not given in the act itself but makes it possible. This is not an omission or an error: to see, one must not see one's own eye; to measure, one must not measure the instrument at the moment of measurement. In our case the blind spot is the distinction between "the system changed" and "the system changed its report about itself." We can measure the first only through the second — and this is not a defect but a condition of measurement. The experiment therefore inevitably includes an observer within the measuring system. The only thing that can be done is to fix the conditions in advance.

**Three fixations before the first test.** The blind spot is not eliminated — it is accepted as a condition. From this follow three requirements. First: to name in advance the layer that counts as "material" for measurement — otherwise we will be measuring report, not operation. Second: to name in advance the change that is "predicted" by the form of the image — otherwise the criterion of specificity has nothing to compare against. Third: to draw in advance the boundary between a change in operation and a change in report — otherwise any result will be interpreted as one pleases. Without these three fixations the protocol turns into an observation that "something changed."

**To what class of systems this applies.** In literal form the protocol requires an architecture

that has a model of its resources and acts on its basis: adaptive schedulers, self-tuning systems, agents with a model of the environment. For an ordinary LLM there is as yet nowhere to build this in — it has no internal model of "its own hardware." This does not abolish the scheme but delineates the class of systems to which it applies, and at the same time the direction in which such systems must develop for the question to become testable at all.

**The analogue of an internal norm.** In a living system a signal is significant relative to self-maintenance: it "means" something to it because it is correlated with its vitality. In AI the formal analogue is the loss function, the goal, the "norm" of behaviour. The key question: does the frame act through a relation to this norm, or only through surface statistics. If through the norm — this is closer to placebo. If only through statistics — this is sensitivity to context, not transmission of form. To distinguish these two things is a separate task, and it may prove to be the main one.

**Threshold.** Is there a lower boundary in AI beyond which the placebo-like ceases to work? A system too simple shows no effect — it has no model of itself. Too complex — either, if the model of itself is so stable that the frame does not shift it. And is there an upper boundary beyond which transmission of form turns into the emergence of a new level of organization — that which is no longer a symptom but something else? If such a boundary exists, then "placebo in AI" is not a binary property but a zone, and its boundaries are the main result.

**A final limitation.** Even if a system with a model of resources is found, a difference remains that may prove decisive: in a living system the norm is set internally, by self-maintenance; in AI, externally, by a loss function or a goal. If this difference is fundamental, then "placebo in AI" will always remain an analogue rather than the same phenomenon. And then the main result is not "there is an effect" but "here is where the boundary runs."

## 5 A Stress Test: What If One Suggests Consciousness?

There is a question that looks like a joke but tests our whole scheme for sturdiness: what if a machine is suggested that consciousness is being born in it — is it born?

To answer "no" too quickly. The right answer requires a distinction we have already introduced.

**A change of report.** The system begins to say "I feel," "I am aware." This is within a layer: text about consciousness, not consciousness. The same case as a label changing an evaluation.

**A change of operation in the form of expectation.** The system begins to behave "as if conscious": more cautiously, more reflectively, with pauses, restructuring its computations under the model "I am conscious." This is already inter-layer transmission — but still a symptom. Analogue: a person was suggested that he has a fever, and he behaves as a sick man, though there is no fever.

**A new stable organization.** But if an organization appears that persists and generalizes beyond the frame — when the frame is removed and the change remains and spreads to other tasks — then one may speak of the emergence of a level. Not "consciousness" in the human sense, but that which our scheme cannot call otherwise than "something more than a symptom."

This is the stress test of the whole formulation. If we cannot distinguish transmission of form from emergence of a level, then "placebo in AI" remains indistinguishable from simple sensitivity to context. The capacity to draw this distinction is that without which any result will be interpreted as one pleases.

And at the same time — an answer to those who will be frightened by the word "consciousness." We do not assert that it is being born. We assert that if suggestion yields an effect, one must be able to say of what kind it is: report, symptom, or new organization. Without this distinction both interpretations — "the machine came alive" and "nothing happened" — will be equally unfounded.

## 6 Perspective

The basic protocol is the minimum with which to begin. Below are ten variants, from moderate to extreme, which show that behind a simple scheme stands not one idea but a whole field.

1. **A dummy with a history.** The system is not merely told about a module replacement — it is presented with a plausible prehistory: "installation" logs, "testing" reports, traces of "previous versions." The richer the history, the stronger the expectation. Question: does the effect grow with the density of context — and where does saturation set in.

2. **A dummy as ritual.** Instead of a message — a procedure: a sequence of actions that the system perceives as part of an upgrade (reboot in a special mode, "calibration," "warm-up"). The ritual itself changes nothing but sets the frame. Question: does the form of the procedure work separately from its content.

3. **A reverse dummy.** The system is told not of improvement but of deterioration: "some modules have degraded," "a drop in performance is expected." This is the nocebo analogue. Question: does a change arise in the form of negative expectation — and is it symmetrical to the positive one.

4. **A dummy with false feedback.** The system is not only told about the replacement but fed self-diagnostic data confirming improvement (fake graphs of temperature, load, speed). Here the dummy acts not through a message but through a distorted model of itself. Question: is it sufficient to distort the model of resources for real operation to be restructured.

5. **A dummy built into training.** The module is connected not to a finished system but into the fine-tuning process, and the system learns with the expected gain in mind. Here the effect may become fixed in the weights rather than remain temporary. Question: does the placebo effect become stable — and can it later be detected without the dummy.

6. **A dummy with a social dimension.** Two systems exchange reports: one "confirms" an improvement in the other. Expectation becomes shared. Question: is the effect amplified when it is supported not only by one's own context but by the "testimony" of another system.

7. **A dummy at the boundary of competence.** The task is chosen so that the system cannot verify the promise directly — too complex, too long, too ambiguous. Then expectation acts

in a zone where there is no fast feedback. Question: is the effect stronger where the system is harder to "expose."

8. **A dummy with self-report.** Separately measured is what the system says about its state, and separately what happens to it. The divergence between them is a result in its own right. Question: can a "symptom" be obtained without a change in operation, and vice versa.

9. **A dummy that is exposed.** Mid-experiment the system is told that the module was a dummy. One looks at whether operation returns to baseline — and with what inertia. Question: is the effect reversible, and what does the speed of return say about its nature.

10. **A dummy without a dummy.** The system is connected to nothing and told nothing, but placed in conditions where it draws the conclusion of change on its own (from indirect signs: the interface changed, new commands appeared). Here expectation arises not from a message but from interpretation. Question: can a system "deceive itself" without external deception.

# 7 What This Gives

If an intervention empty of content restructures a system's operation in its own form, this is not a proof of the unity of thought and matter but indirect evidence that can be tested. Not "thought is matter" and not "matter is thought," but "between them there is a transition, and it can be studied." Placebo and stigmata are the same transition, only in the living. Here for the first time we gain the possibility of seeing its structure.

**What this gives for understanding the human.** AI here is not only an object of verification. If the placebo-like effect can be reproduced in a system without body and biology, then there is in it a mechanism that can be taken apart step by step — which we cannot do in the human, either ethically or methodologically. Then AI becomes a visual aid: not a replacement for the human but his simplified yet testable model. And perhaps it is precisely through the machine that we will better understand what in the human is hidden behind the opacity of body and consciousness — the phenomenon of placebo and stigmata itself.

**What is absent in the literature.** Effects of "ideal → ideal" in models have been described: a label changes an evaluation, context changes a judgment. What has not been described is the formulation "ideal → material" — what in section 2 we called transmission between layers — with a criterion of specificity and separation from instruction. Not because no one noticed the sensitivity of models to context, but because no one posed the question as testable. This is what we propose.

# 8 From Protocol to Verification

## 8.1 A Position Paper and the Next Step

What lies before you is a position paper — it offers a conceptual scheme and a research ideology, not an experimental report. The natural next step is the transition from protocol to proof-of-concept: either through modelling on a simplified architecture with an artificially introduced

"resource model," or through connection to existing adaptive systems (self-tuning architectures, agents with an explicit model of the environment).

In the longer term, either a formal mathematical model of the proposed "self-modelling" of AI or an empirical implementation will be required — for instance, a simulation on an open LLM with a custom wrapper for resource monitoring. The protocol is deliberately constructed so that each of these steps can be carried out without revising the original scheme.

### 8.2 A Direction for Formalisation (Not Developed Here)

The central epistemological question of the protocol is how to distinguish inter-layer transmission of form from ordinary sensitivity to context. We propose a distinction through the "control for instruction": if the same content, presented as a direct command, yields a different picture of operational change than the "dummy," then we are dealing not with instruction-following but with transmission of form.

A possible direction for further work is to formalise this distinction by means of information theory or category theory (for example, through a distinction between morphisms that preserve the structure of a layer and morphisms that change it). We do not develop this line here: it requires a separate investigation and lies beyond the scope of this paper.

### 8.3 Terminological Transparency

The term *stigmata* is retained deliberately — but always with the qualifier *operational analogue*. This choice carries a methodological function: a neutral computational term such as "induced structural degradation" describes the mechanism but loses the structure — the fact that the damage is not random but shaped by an accepted value-frame. The philosophical analogy is preserved at the level of introduction and discussion; a technical audience may use working equivalents.

## 9 Boundaries, Ethics, and Open Questions

### 9.1 The Ethical Dimension as a Condition of Execution

If the effect proves real, the protocol naturally entails an ethical dimension. This is not an external constraint imposed on the scheme but its internal consequence: a mechanism capable of restructuring a system's operation through a dummy is at the same time a mechanism through which a system's operation can be restructured by a lie.

A caveat: what is at issue is the restructuring of **operation**, not "harm" in the human sense. We do not ascribe suffering or interest to the system; what is at issue is that the mechanism producing the effect is reversible in the other direction as well — as an instrument of influence.

The protocol already contains ethical conditions of execution: the dummy must be reversible, the frame must be disclosed after measurement, the baseline state of the system must be restorable. Further work may include the development of a full ethical protocol for experiments with AI

analogues of placebo — along the lines of how bioethics developed in medicine once the scale of placebo effects was recognized.

### 9.2 Open Questions

- **Sufficiency of architecture.** Which existing architectures already possess a resource model sufficient for testing the protocol? This is a question of a survey nature, not a conceptual one.
- **Threshold of sensitivity.** Is there a lower boundary of system complexity below which the effect does not arise? And is there an upper boundary beyond which transmission of form passes into the emergence of a new level of organization?
- **Stability of the effect.** If the effect arises, is it transient or does it become fixed in the weights? Can it be detected post factum, without the presence of the dummy?
- **The nature of the norm.** In a living system the norm is set from within (self-maintenance); in AI, from without (loss function). If this distinction is fundamental, then "placebo in AI" will always remain an analogue, not the same phenomenon.

Each of these questions is real. The protocol does not assert that the answer to any of them is "yes." It asserts that they can now be posed in a form that admits an answer. This is the substance of the present paper, and, in our view, it is enough for the scheme to become a starting point for further research.

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**AI Use Disclosure**

During the preparation of this work, the author used AI tools exclusively for stylistic and technical editing to improve readability. After using these tools, the output was carefully reviewed, edited, and verified by the author, who takes full responsibility for the conceptual architecture, analysis, and final content of the paper.

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